llmXive Automated Review of Video-Oasis: Rethinking Evaluation of Video Understanding

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong claims regarding the prevalence of "shortcuts" in video benchmarks and the resulting performance of state-of-the-art models. While the methodology for auditing benchmarks is sound, there are specific instances where the textual claims are not fully supported by the provided evidence or rely on unverifiable external resources.

First, the claim in Section 4.2 that models perform "marginally above random guessing" is an overstatement given the data. The text cites a random chance baseline of 25.6% and claims models are only slightly above this. However, Table 7 shows the lowest-performing open-source model (Video-R1) achieves 26.3%. A 0.7% margin is statistically fragile and arguably indistinguishable from noise without a significance test, making the phrase "marginally above" misleading. The authors should either soften this claim to "near random chance" or provide statistical evidence that the 0.7% gap is significant.

Second, the reliance on "GPT-5" and "o4-mini" for the consensus labeling in Section 4.1 is problematic. The bibliography lists "GPT-5" as a 2026 publication. As of the current review date, GPT-5 has not been publicly released or documented in a verifiable system card. Using a non-existent or unreleased model as a primary instrument for filtering the dataset (identifying the 122 samples requiring manual inspection) invalidates the reproducibility of the "video-native" challenge set. The authors must verify the actual model used (e.g., GPT-4o) and correct the citation, or explicitly state that these are hypothetical baselines, which would weaken the claim of a rigorous audit.

Finally, there is a minor numerical discrepancy in Section 3.2. The text states an average shortcut ratio of 92.7% under $c \geq 1$, but the arithmetic mean of the four values in Table 3 (95.6, 95.7, 85.8, 94.0) is 92.775%, which rounds to 92.8%. Given the paper's focus on precise auditing, this rounding error should be corrected. Additionally, Section 5.3 contains a LaTeX truncation error that cuts off the conclusion regarding RLVR reward designs, preventing the reader from verifying the authors' interpretation of the data.

These issues do not invalidate the core contribution of the Video-Oasis framework but require correction to ensure the claims are rigorously supported by the evidence presented.

Action items.

- [**writing**] The paper makes several strong claims regarding the prevalence of "shortcuts" in video benchmarks and the resulting performance of state-of-the-art models. While the methodology for auditing benchmarks is sound, there are specific instances where the textual claims are not fully supported by the provided evidence or rely on unverifiable external resources. First, the claim in Section 4.2 that models perform "marginally above random guessing" is an overstatement given the data. The text cites a r

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`
Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively illustrates the existence of shortcuts in panel (a) and the performance gap in panel (c), but panel (b) presents a confusing negative correlation that contradicts the caption's claim of a positive relationship between shortcut ratios and accuracy.

Figure 2

Figure 2 is a clear and well-structured schematic that effectively communicates the three components of the Video-Oasis diagnostic suite. The visual flow from input to specific tests and final agreement metrics is logical, and the caption accurately describes the figure's content.

Figure 3

The figure presents qualitative examples of video QA failures, but the caption for panel (b) contradicts the visual evidence by claiming temporal questions were 'incorrectly filtered' when they clearly demonstrate the temporal reasoning the filter is meant to catch.

Figure 4

Figure 4 effectively visualizes the distinction between previous benchmarks containing shortcuts and the video-native challenges distilled by Video-Oasis. The layout is clear, the legend defining the checkmarks and crosses is present, and the qualitative examples align perfectly with the caption's description of the five challenge categories.

Figure 5

The figure provides clear visual examples of video QA tasks with bounding boxes and text, but the internal labels ('Living Room Recognition', 'Background Tracking') contradict the caption's classification of 'Fine-Grained Perception Challenges,' and the layout contains redundant headers.

Figure 6

Figure 6 provides clear qualitative examples of spatial world understanding challenges, featuring distinct video frames, specific navigation questions with fill-in-the-blank prompts, and a breakdown of the required reasoning steps. The layout is uncluttered, and the text is legible, effectively supporting the caption's description.

Figure 7

Figure 7 provides clear qualitative examples of temporal dynamics and tracking challenges using video frames with bounding boxes and descriptive text. The visual layout effectively illustrates the specific tasks (Fish Tracking, Location Linking, Order Reconstruction, etc.) without requiring additional legends or controls.

Figure 8

Figure 8 provides clear qualitative examples of causality and logical reasoning challenges, effectively illustrating the 'Video Native Challenges' workflow described in the caption. The visual layout, including the video frames, bounding boxes, and the step-by-step reasoning breakdown, is legible and well-structured.

Figure 9

Figure 9 provides clear qualitative examples of Global Narrative Challenges, using video frames with bounding boxes to highlight specific events and text to explain the temporal reasoning required. The layout is uncluttered, and the connection between the visual evidence and the narrative question is explicit.

Action items.

- [**science**] Figure 1b: The caption claims 'Benchmarks with higher shortcut ratios tend to report higher accuracy,' but the chart shows a negative correlation where accuracy (blue bars) generally decreases as the shortcut ratio (red line) decreases. The trend contradicts the caption's claim.
- [**writing**] Figure 1b: The x-axis labels for the benchmarks are rotated at a steep angle, making them difficult to read and cluttered.
- [**science**] Figure 3: The caption claims panel (b) shows 'Questions incorrectly filtered by the frame shuffling test,' but the visual examples (baseball, basketball) explicitly highlight 'Temporal Context' and sequential events ('after the second ball'), which are the exact type of temporal dependencies the frame shuffling test is designed to detect. This contradicts the caption's assertion that these were 'incorrectly filtered' and should have been removed.
- [**writing**] Figure 3: Panel (a) bottom example uses the label 'Ambiguous Subject' to describe a scene with multiple bounding boxes, but the visual evidence (red vs. green boxes) suggests the issue is 'Object Tracking' or 'Identity Consistency' rather than subject ambiguity, as the question asks for a relation between two specific entities.
- [**writing**] Figure 5: The caption states this figure shows 'Fine-Grained Perception Challenges,' but the internal text labels explicitly categorize the examples as 'Living Room Recognition' and 'Background Tracking' (Spatial Temporal Grounding), creating a mismatch between the figure's title and its content.
- [**writing**] Figure 5: The text 'Why Video Native Challenges?' is repeated as a header for both examples, which is redundant and disrupts the visual flow of the qualitative examples.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent researcher in adjacent fields, with most standard acronyms (e.g., Video-LLM, SFT) either well-known or defined upon first use. However, there are specific instances where subfield-specific acronyms or method names are introduced without explicit expansion, creating minor barriers for an outsider.

In Section 5.1, the authors refer to "AKS" to denote the method from the cited paper "Adaptive keyframe sampling for long video understanding." The acronym is used without first spelling out "Adaptive Keyframe Sampling (AKS)." While the citation provides the source, the text itself assumes prior knowledge of the acronym, which violates the self-containment requirement for an adjacent-field reader.

Additionally, in Section 3.3, the terms "sensitivity check" and "redundancy checks" are used in lowercase. These refer to the specific "Sensitivity" and "Redundancy" tests defined in Section 3.1. Without capitalization or a clarifying reference (e.g., "the Sensitivity test defined in Section 3.1"), a reader might interpret these as generic statistical checks rather than the specific protocols of the Video-Oasis framework.

Finally, in Section 5.2, the "Qwen3-VL_voting" baseline is described as an "oracle ensemble," but the specific aggregation logic (e.g., logical OR vs. majority vote) is not explicitly stated in the definition sentence, relying on the reader to infer the mechanism from the following sentence. Explicitly stating the rule (e.g., "accepts a response if either mode succeeds") would remove this ambiguity.

These issues are minor and can be resolved with simple parenthetical expansions or clarifying phrases, ensuring the paper remains fully self-contained.

Action items.

- [**writing**] Section 5.1 uses 'AKS' without expansion. Define as 'Adaptive Keyframe Sampling (AKS)' at first use.
- [**writing**] Section 3.3 refers to 'sensitivity check' and 'redundancy checks' without linking to the capitalized definitions in 3.1. Clarify as 'the Sensitivity test' and 'the Redundancy test' to distinguish from generic terms.
- [**writing**] Section 5.2 introduces 'Qwen3-VL_voting' as an 'oracle ensemble baseline' without defining the aggregation rule. Specify 'logical OR' or 'majority vote' explicitly in the definition clause.
- [**writing**] Section 3.1 introduces 'Bag-of-Frames (BoF)' but the acronym 'BoF' is used immediately. Ensure the full expansion 'Bag-of-Frames (BoF)' is clearly presented before the acronym is used in subsequent sentences.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument structure: identifying a problem (benchmark shortcuts), proposing a solution (Video-Oasis), and validating it. However, two specific logical gaps prevent the conclusions from fully following the premises as written.

First, **Section 5.3 (Training Paradigms)** contains a critical logical break due to a text truncation error. The section introduces an analysis of SFT vs. RLVR and lists key insights. The first insight is complete, but the second (“RLVR Reward Designs”) cuts off mid-sentence: “Video-R1 (26.3%) } \\ \\texttt{< Definition > : A clear academic definition...”. The text immediately jumps to a prompt template, omitting the actual finding regarding RLVR reward designs. Consequently, the paper’s claim to have derived “practical guidelines” for algorithmic design is unsupported in this section because the argument is incomplete. The conclusion regarding RLVR does not follow because the premise (the result of the comparison) is missing.

Second, there is a **scope inflation** between the Abstract and the body results regarding the “55%” shortcut prevalence. The Abstract states definitively: “This audit reveals that 55% of existing benchmark samples are solvable without visual input or temporal context.” In Section 3.2, Table 3 presents shortcut ratios under three different consensus thresholds ($c \geq 1$, $c \geq 2$, $c = 3$). The values for $c = 3$ range from 44.6% to 63.0% across categories. The paper does not explicitly state which calculation yields the “55%” figure (e.g., is it the unweighted average of the $c = 3$ row? A weighted average?). Without this explicit derivation in the body, the specific number “55%” in the Abstract appears to be an unsupported specific conclusion drawn from a range of data. To ensure the conclusion follows logically, the body must explicitly define how the aggregate 55% was calculated or qualify the Abstract’s claim.

Action items.

- [**science**] Section 5.3 text is truncated mid-sentence (‘Video-R1 (26.3%) } \ \texttt{< Definition...’), cutting off the RLVR analysis. The conclusion about RLVR effectiveness does not follow from the missing premise. Complete the sentence and state the finding derived from Table 11.
- [**writing**] Abstract claims ‘55% of samples are solvable without visual/temporal input,’ but Section 3.2 Table 3 shows a range (44.6%-63.0%) for strict consensus (c=3). The body does not define how the specific ‘55%’ aggregate was calculated. Explicitly state the calculation method or qualify the Abstract claim to match the data.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a rigorous diagnostic framework, but the rhetoric in the abstract and conclusion occasionally exceeds the specific scope of the evidence provided in the tables.

First, the abstract highlights that “55% of existing benchmark samples are solvable without visual input.” This figure is derived from the strict consensus threshold ($c = 3$) in Table 5. However, Table 3 demonstrates that under a relaxed threshold ($c \geq 1$), the shortcut prevalence rises to 92.7%. By presenting the conservative lower bound as the primary finding without qualification, the abstract risks understating the severity of the problem for readers who do not immediately cross-reference the detailed tables. The claim should be contextualized to reflect that the “55%” figure represents a strict consensus estimate, or the abstract should acknowledge the significantly higher prevalence under relaxed conditions.

Second, the abstract and Section 4.2 assert that state-of-the-art models perform “only marginally above random guessing.” This generalization is factually inaccurate when applied to the entire set of evaluated models. Table 7 clearly shows that the frontier model, Gemini-2.5-Pro, achieves 46.7% accuracy, which is nearly double the 25.6% random baseline. While the claim accurately describes the performance of open-source models (which hover around 26-36%), it incorrectly implies that *all* state-of-the-art models are failing at random levels. The text should explicitly scope this observation to “open-source Video-LLMs” to avoid misrepresenting the capabilities of the leading proprietary models.

Third, the conclusion in Section 5.2 claims that “strategic optimization of when to think can be as impactful as the raw scale of the model’s architecture.” This comparison is not supported by the experimental design. The “oracle ensemble” experiment compares a specific Qwen3-VL configuration against itself and a proprietary model, but it does not test “raw scale” across a range of model sizes (e.g., 7B vs 70B) to support a general claim about scaling laws. The claim should be narrowed to state that optimizing reasoning depth can bridge the performance gap between current open-source models and frontier proprietary models, rather than equating strategy with scaling.

Action items.

- [**writing**] Abstract: The ‘55%’ shortcut claim relies on a strict consensus threshold (c=3) in Table 5, while Table 3 shows 92.7% under relaxed conditions (c>=1). Scope the claim to ‘under strict consensus’ or acknowledge the higher prevalence to avoid understating the issue.
- [**writing**] Abstract & Sec 4.2: Claiming SOTA models perform ‘marginally above random’ is inaccurate for Gemini-2.5-Pro (46.7% vs 25.6% baseline). Scope this to ‘open-source Video-LLMs’ to avoid misrepresenting frontier proprietary model capabilities.
- [**writing**] Sec 5.2: The claim that reasoning strategy is ‘as impactful as raw scale’ lacks a scaling study. The oracle ensemble only compares modes within one model. Narrow to ‘can close the gap between current open-source and frontier models’.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a diagnostic framework (Video-Oasis) for auditing existing video understanding benchmarks to identify "shortcut" samples that can be solved without visual or temporal evidence. The work involves re-evaluating public benchmarks (e.g., EgoSchema, VideoMME, MLVU) and running automated tests (e.g., blind tests, frame shuffling) on them.

From a safety and ethics perspective, the research is low-risk. The methodology does not involve:

1. **Human Subjects:** The "human-in-the-loop" verification mentioned (Section 3.1, Criteria 3) refers to inspecting annotation quality of existing public datasets, not collecting new data from human participants. No IRB/ethics statement is required for this type of secondary analysis of public data.
2. **Dual-Use Harm:** The paper does not generate new capabilities for deception, surveillance, or cyber-attacks. Instead, it aims to *reduce* the overestimation of model capabilities, which is a safety-positive outcome for the field.
3. **Data Privacy:** The work relies on existing public benchmarks. There is no indication of scraping private data, releasing PII, or violating data licenses in the context of the *new* artifacts produced (the filtered set and the diagnostic code). The authors explicitly state code availability and the nature of the distilled dataset.
4. **Bias/Fairness:** While the paper discusses benchmark flaws, it does not introduce new biases or fail to address a specific, identifiable harm to a demographic group that the paper's own results expose.

The paper appropriately focuses on methodological rigor in evaluation rather than deploying systems with direct societal impact. No specific safety disclosures or mitigations are missing given the nature of the work. The verdict is **accept** with no action items required for this lens.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling diagnostic framework, Video-Oasis, to audit video benchmarks. However, the evidentiary strength of the core quantitative claims is weakened by a lack of reported variance and insufficient controls in the ablation studies.

First, the headline finding that “55% of existing benchmark samples are solvable without visual input” (Abstract) and the detailed breakdowns in Table 2 and Table 3 are presented as aggregate point estimates. The paper does not report standard deviations, confidence intervals, or the number of random seeds used for the diagnostic models (Eagle2.5, Qwen2.5-VL, etc.). In large-scale benchmark evaluations, model performance can vary significantly across seeds or minor hyperparameter shifts. Without reporting variance (e.g., mean $\pm$ SD over 3-5 seeds), the reader cannot determine if the reported shortcut ratios are stable phenomena or artifacts of a specific model initialization. The claim that “state-of-the-art models perform only marginally above random guessing” on the distilled set (Table 7) similarly lacks variance metrics, making it difficult to assess the statistical significance of the gap between models.

Second, the ablation study on temporal grounding in Section 5.1 (Table 8) introduces a potential confound. The authors report modest accuracy gains (1.4% to 2.3%) when adding the AKS temporal grounding module. However, the AKS module involves an additional retrieval step that likely increases inference time and computational cost compared to the baseline. The design does not control for this “compute budget” difference. It is plausible that the performance gain arises simply from the model having access to more processed information or a longer inference window, rather than the specific logic of the AKS grounding. To isolate the contribution of the grounding mechanism, the authors should run a control experiment where the baseline model is given an equivalent amount of compute or retrieved frames (e.g., via a random or naive retrieval strategy) to ensure the gain is not merely a function of increased resources.

Finally, the categorization of video-native challenges (Section 4.1) relies on an ensemble of proprietary LLMs (GPT-4o, GPT-5, etc.) to label the filtered samples. While the authors mention a consensus threshold, they do not report the inter-annotator agreement (e.g., Cohen’s Kappa) or the variance in category distribution if different model ensembles were used. Given that the subsequent evaluation (Table 7) is broken down by these categories, the stability of these labels is critical. If the categorization is noisy, the observed performance differences across categories (e.g., “Global Narrative” being the hardest) could be an artifact of label inconsistency rather than a true capability gap.

Addressing these issues by reporting variance metrics for the diagnostic results and adding a compute-matched control for the grounding ablation would significantly strengthen the scientific validity of the paper’s conclusions.

Action items.

- [**writing**] The paper presents a compelling diagnostic framework, Video-Oasis, to audit video benchmarks. However, the evidentiary strength of the core quantitative claims is weakened by a lack of reported variance and insufficient controls in the ablation studies. First, the headline finding that “55% of existing benchmark samples are solvable without visual input” (Abstract) and the detailed breakdowns in Table 2 and Table 3 are presented as aggregate point estimates. The paper does not report standard de

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment in this paper is generally descriptive, which is common for benchmark auditing papers, but it lacks necessary rigor in quantifying uncertainty and validating baselines.

First, the paper relies heavily on aggregate accuracy percentages (e.g., "55% of samples," "92.7% shortcut ratio," "35.6% accuracy") presented as precise point estimates in Tables 2, 3, 4, 5, 6, 7, 8, 9, and 10. There is no reporting of standard deviation (SD), standard error (SE), or confidence intervals (CI) for these aggregates. Since these numbers are derived from aggregating results across 14 different benchmarks and multiple models, the variance across these sources is unknown. A reader cannot determine if a reported "92.7%" is a stable finding or an artifact of a specific subset of benchmarks. The authors should report the standard deviation across the 14 benchmarks for the aggregate metrics to demonstrate the robustness of the "shortcut prevalence" claim.

Second, the "random-chance baseline" of 25.6% cited in Section 4.2 and used to interpret model performance as "marginally above random" is not explicitly derived. This number implies an average of approximately 3.9 options per question (1/0.256). However, the paper does not provide the distribution of multiple-choice options (e.g., how many questions have 3, 4, or 5 choices) across the 14 audited benchmarks. If the option distribution is skewed, the true random baseline could differ significantly. The authors must explicitly state the option distribution or calculate the weighted random baseline based on the actual dataset composition to ensure the "random chance" comparison is valid.

Third, in Section 5.1 (Table 8), the paper claims that temporal grounding yields "consistent improvements" (e.g., Eagle2.5: 31.5% → 32.9%). These are presented as single point estimates without any statistical test. Since the same test set is used for both the baseline and the ablation, the difference should be evaluated using a paired statistical test (such as McNemar's test for classification accuracy or a paired bootstrap) to determine if the improvement is statistically significant or within the noise of the evaluation. Reporting a p-value or a confidence interval for the difference is necessary to support the claim of improvement.

Finally, while the paper uses an ensemble of models for categorization (Section 4.1), it does not report the inter-annotator agreement (e.g., Cohen's Kappa) for the 122 manually inspected samples or the consensus rate for the LLM ensemble. Given the reliance on these categories for the final evaluation split, a measure of annotation reliability is missing.

Action items.

- [**writing**] The statistical treatment in this paper is generally descriptive, which is common for benchmark auditing papers, but it lacks necessary rigor in quantifying uncertainty and validating baselines. First, the paper relies heavily on aggregate accuracy percentages (e.g., "55% of samples," "92.7% shortcut ratio," "35.6% accuracy") presented as precise point estimates in Tables 2, 3, 4, 5, 6, 7, 8, 9, and 10. There is no reporting of standard deviation (SD), standard error (SE), or confidence interval

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the prose is clear, allowing the reader to follow the argument from the problem statement to the proposed diagnostic suite and results. The abstract effectively summarizes the method and key findings. However, there are a few specific instances where the flow is disrupted by incomplete sentences or minor grammatical inconsistencies that require attention.

The most critical issue is in Section 5.3 ("Training Paradigms"). The paragraph introducing the key insights regarding RLVR reward designs ends abruptly. The sentence "The comparative results among RLVR-based models and the SFT baseline reveal a non-linear performance trend: Video-R1 (26.3%) }" cuts off mid-thought, followed by a stray LaTeX brace and a fragment. This breaks the reader's momentum and obscures the intended conclusion about the effectiveness of specific reward designs. The authors must complete this sentence to ensure the argument is fully articulated.

Additionally, there is a minor inconsistency in subject-verb agreement regarding the word "data." In Section 3.1, the text states, "Since video data are long and information-dense..." treating "data" as a plural noun. While this is technically correct in strict academic English, the paper occasionally shifts toward treating "data" as a singular mass noun in other contexts (e.g., "the data is..."). For a formal publication, it is best to choose one convention and apply it consistently throughout the manuscript to avoid distracting the reader.

Finally, in Section 4.1, the sentence describing the consolidation of clusters is slightly verbose. While grammatically correct, tightening the phrasing (e.g., removing "that capture" or "initial") would improve the rhythm of the paragraph without losing meaning. Addressing these specific points will ensure the paper reads as smoothly as its scientific content warrants.

Action items.

- [**writing**] Section 5.3, paragraph 2: The sentence listing RLVR insights ends abruptly with 'Video-R1 (26.3%) }' and is followed by a LaTeX fragment, cutting off the conclusion. Complete the sentence to state the finding regarding reward design effectiveness before the list ends.
- [**writing**] Section 3.1, paragraph 3: The sentence 'Since video data are long and information-dense, video-QA annotations can be ambiguous...' uses 'are' with the singular subject 'data' (often treated as singular in modern usage, but 'data' is plural in strict academic style). For consistency with the rest of the paper's formal tone, consider 'Since video data is long...' or

'Since video data are long...' (ensure consistency with other instances of 'data' in the text).

- **[writing]** Section 4.1, paragraph 2: The phrase 'We then consolidate these initial clusters into five unified categories that capture the dominant capabilities required by the Video-Oasis-filtered samples' is slightly wordy. Consider tightening to 'We consolidate these into five categories capturing the dominant capabilities required by Video-Oasis-filtered samples.' to improve flow.